

Municipal OTMS - Demand & Collection SQL Reports

This document contains SQL queries for generating municipal tax demand and collection reports for House Tax, Water Tax, Sewer Tax, and Mutation-related operations.

1. Total Demand by Ward

```
SELECT
    w.ward_name,
    SUM(t.total_tax) AS total_demand
FROM Tax_Assessment t
JOIN Property p
    ON t.property_id = p.property_id
JOIN Ward w
    ON p.ward_id = w.ward_id
GROUP BY w.ward_name;
```

2. Total Collection by Ward

```
SELECT
    w.ward_name,
    SUM(tp.paid_amount) AS total_collection
FROM Tax_Payment tp
JOIN Tax_Assessment ta
    ON tp.assessment_id = ta.assessment_id
JOIN Property p
    ON ta.property_id = p.property_id
JOIN Ward w
    ON p.ward_id = w.ward_id
WHERE tp.payment_status = 'PAID'
GROUP BY w.ward_name;
```

3. Demand and Collection by Date Range

```
SELECT
    w.ward_name,
    SUM(ta.total_tax) AS total_demand,
    SUM(tp.paid_amount) AS total_collection
FROM Ward w
JOIN Property p
    ON w.ward_id = p.ward_id
JOIN Tax_Assessment ta
    ON p.property_id = ta.property_id
LEFT JOIN Tax_Payment tp
    ON ta.assessment_id = tp.assessment_id
WHERE ta.assessment_date
    BETWEEN '2026-04-01' AND '2026-04-30'
GROUP BY w.ward_name;
```

4. Pending Balance Report

```
SELECT
    p.property_number,
    o.full_name,
    ta.total_tax,
    COALESCE(SUM(tp.paid_amount), 0) AS total_paid,
    (
        ta.total_tax -
        COALESCE(SUM(tp.paid_amount), 0)
    ) AS pending_balance
```

```

    ) AS balance_due
FROM Tax_Assessment ta
JOIN Property p
    ON ta.property_id = p.property_id
JOIN Property_Owner po
    ON p.property_id = po.property_id
JOIN Owner o
    ON po.owner_id = o.owner_id
LEFT JOIN Tax_Payment tp
    ON ta.assessment_id = tp.assessment_id
GROUP BY
    ta.assessment_id,
    p.property_number,
    o.full_name,
    ta.total_tax;

```

5. Daily Collection Report

```

SELECT
    payment_date,
    SUM(paid_amount) AS daily_collection
FROM Tax_Payment
WHERE payment_status = 'PAID'
GROUP BY payment_date
ORDER BY payment_date;

```

6. Ward-wise Recovery Percentage

```

SELECT
    w.ward_name,
    SUM(ta.total_tax) AS demand,
    SUM(tp.paid_amount) AS collection,
    ROUND(
        (
            SUM(tp.paid_amount) /
            SUM(ta.total_tax)
        ) * 100,
        2
    ) AS recovery_percentage
FROM Ward w
JOIN Property p
    ON w.ward_id = p.ward_id
JOIN Tax_Assessment ta
    ON p.property_id = ta.property_id
LEFT JOIN Tax_Payment tp
    ON ta.assessment_id = tp.assessment_id
GROUP BY w.ward_name;

```

Best Practices

Recommended Additional Fields:

Tax_Assessment:

- demand_status
- generated_date
- due_date

Tax_Payment:

- created_at
- updated_at
- payment_source